AI Exposure or Post-Pandemic Employment Adjustment? Explaining the Decline in Early-Career Hiring

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Research Question and Study Design

This study will examine the relationship between generative artificial intelligence (AI) exposure and early-career hiring after 2022. The main research question is: After accounting for employment growth above pre-COVID trends, is greater generative AI exposure associated with a larger decline in hiring among workers ages 22 to 24? Previous research has found weaker early-career hiring in industries with greater AI exposure [1]. However, some of these industries also experienced unusual employment growth during the pandemic, which may help explain their later hiring decline [2].

This will be a non-experimental, observational study using publicly available data. I will compare hiring patterns across industries and states with different levels of AI exposure and employment above their pre-COVID trends. The main statistical approach will be a fixed-effects panel regression that compares changes in early-career hiring before and after 2022 across state-industry groups. I will also estimate an event-study specification to examine how these differences developed quarter by quarter. These methods will help account for stable differences across state-industry groups and economy-wide changes over time, but they will not remove every possible explanation for the observed changes. Because AI exposure was not randomly assigned and other economic changes occurred during the same period, I will interpret the estimates as associations rather than definitive causal effects.

Data Sources and Study Population

The main dataset will be the U.S. Census Bureau's Quarterly Workforce Indicators (QWI), which provide quarterly information on employment, hiring, and worker separations [3]. The QWI extraction will cover 2015 Q1 through the latest quarter shared across all included state-age groups. The sample will include all usable states and the District of Columbia, four-digit NAICS industries, private-sector employment, all sexes, and unadjusted quarterly data. The main age group will be workers ages 22 to 24, while the all-ages category will be used to construct the employment-above-trend measure. Each observation in the working data will represent a state, four-digit industry, age group, and quarter. Alaska will be excluded because its available QWI series ends in 2016 Q2, and Michigan will be excluded because its series ends in 2021 Q4. Neither state therefore provides usable post-2022 observations for the main research outcome.

I will combine QWI with occupation-level AI exposure scores from Eloundou et al. [4]. I will use their existing GPT-4 beta exposure measure rather than create new ratings. Because the exposure scores are organized by occupation, I will use pre-pandemic American Community Survey Public Use Microdata Sample (ACS PUMS) employment data to calculate the share of workers in each industry employed in different occupations [5]. I will then use these shares to calculate a national AI exposure score for each industry. I will match occupation and industry codes using Census classification guides and crosswalks [6]. Additionally, I will assign the same industry exposure score to that industry across states and quarters.

The QWI and ACS PUMS are publicly available Census datasets, and the authors provide the occupation-level AI exposure scores through their public research repository. I will follow the applicable data-use and sharing requirements and will not collect personally identifiable information. I will retain the original Census API responses, variable definitions, and status flags separately from the cleaned analytical data. Within the included states, I will require each state-industry group to contain at least 16 of the 20 quarterly employment observations from 2015 Q1 through 2019 Q4. This threshold provides at least 80% coverage of the pre-pandemic period while allowing for a small number of Census-suppressed observations. The available observations must also be distributed across the pre-pandemic years and quarters so that a time trend and seasonal effects can be estimated. State-industry groups must also have usable all-age employment observations in 2022 Q1 through Q3 and usable early-career hiring observations before and after 2023. Missing and suppressed observations will remain missing rather than being treated as zero. I will compare the main sample with a stricter sample containing all 20 pre-pandemic quarters as a sensitivity check.

Variables and Measurement

The main dependent variable will be the Census-defined quarterly hiring rate for workers ages 22 to 24 in each state-industry group. I will use the QWI variable definitions retained by the data pipeline to report the exact Census numerator and denominator. I will analyze the number of hires as a secondary outcome because a hiring rate may change partly because the size of the employed workforce changes. I will also examine employment and worker separations as secondary outcomes to distinguish changes in hiring from changes in employment or worker exits.

The first main independent variable will be industry-level exposure to generative AI. I will calculate this score using the GPT-4 occupation-level measure developed by Eloundou et al. [4], weighted by each occupation's share of employment within an industry before the pandemic. A higher score will indicate that a larger share of the industry's work tasks could potentially be affected by generative AI. It will not measure how much AI firms actually adopted or whether workers were replaced.

The second main independent variable will measure employment above the pre-COVID trend. For each state-industry group, I will use total employment data from 2015 through 2019 to estimate the employment level expected in 2022 if its earlier trend had continued. I will compare predicted employment with actual employment during 2022 Q1 through Q3 and calculate the average percentage difference. A positive value indicates employment was above its predicted pre-COVID trend, while a negative value indicates it was below trend. I will refer to this variable as employment above the pre-COVID predicted trend rather than overhiring because it does not show whether firms hired too many workers or whether the pandemic caused the difference.

I will estimate expected employment using a linear regression fitted separately to each state-industry group's quarterly all-age employment from 2015 through 2019. The model will include a linear time trend and quarter indicators to account for regular seasonal differences in employment. I will use a linear specification as the primary model because it provides a transparent and parsimonious baseline with only 20 pre-pandemic quarterly observations per group. However, a linear trend may not fit industries that experienced accelerating or other nonlinear growth before COVID. I will therefore inspect the pre-pandemic employment trends and test the sensitivity of the employment-above-trend measure using an alternative starting year and a log-linear trend specification [7]. If an industry's estimated employment deviation changes substantially when the

starting year or functional form changes, I will treat its above-trend measure as less stable and interpret the corresponding regression results cautiously.

Statistical Analysis

I will begin with descriptive statistics and graphs showing early-career hiring trends across industries with different levels of AI exposure and employment above their pre-COVID predicted trends. I will then estimate the following fixed-effects panel regression [11]:

$$HiringRate_{ist} = \alpha_{is} + \lambda_t + \beta_1(AIExposure_i \times Post_t) + \beta_2(AboveTrend_{is} \times Post_t) + \epsilon_{ist}$$

In this model, i represents industry, s represents state, and t represents quarter. $HiringRate_{ist}$ is the hiring rate for workers ages 22 to 24. $AIExposure_i$ is the national industry-level generative AI exposure score. $AboveTrend_{is}$ is the percentage by which a state-industry group's 2022 Q1 through Q3 employment was above or below its predicted pre-COVID trend. $Post_t$ equals zero for quarters from 2015 Q1 through 2022 Q3 and one beginning in 2023 Q1. I will exclude 2022 Q4 as a transition quarter. State-industry fixed effects, α_{is}, will account for stable differences across state-industry groups, and quarter fixed effects, λ_t, will account for changes shared across groups in the same quarter. The coefficient β_1 will show whether greater AI exposure is associated with a different post-2022 change in early-career hiring after accounting for employment above trend. The coefficient β_2 will show whether employment above trend is associated with a different post-2022 hiring change. These estimates will be interpreted as conditional associations, not causal effects.

As a secondary analysis, I will estimate an event-study specification that replaces the single post-period interaction with quarter-specific interactions:

$$HiringRate_{ist} = \alpha_{is} + \lambda_t + \sum_{q \neq 2022Q3} \beta_q(AIExposure_i \times Quarter_t) + \sum_{q \neq 2022Q3} \theta_q(AboveTrend_{is} \times Quater_{qt}) + \epsilon_{ist}$$

I will use 2022 Q3 as the reference quarter and exclude 2022 Q4 as a transition quarter. The β_q coefficients will show how the relationship between AI exposure and early-career hiring in each quarter differs from the relationship in 2022 Q3. The θ_q coefficients will provide the corresponding comparisons for employment above the pre-COVID trend. I will examine the coefficients before 2023 to determine whether industries with different levels of AI exposure were already following different hiring patterns before ChatGPT's release. Evidence of earlier differences would make a post-2022 AI interpretation less convincing [8].

Model Validation and Limitations

I will evaluate the results' reliability using several sensitivity checks. First, I will examine the pre-2023 event-study estimates to determine whether industries with different levels of AI exposure were already following different hiring patterns before ChatGPT's release. Second, I will test whether the employment-above-trend measure depends strongly on the selected pre-pandemic trend

specification. I will re-estimate the measure using a pre-pandemic window beginning in 2016 rather than 2015 and will compare the linear trend with a log-linear trend. Large changes across these specifications would indicate that the employment-above-trend measure is sensitive to modeling choices. Third, I will compare the main sample with a complete-case sample containing all 20 pre-pandemic quarters. I will also examine whether the estimated relationships change after excluding major technology industries and after accounting for industry-level remote-work exposure, following Dingel and Neiman [9].

I will report the number of states, industries, quarters, and usable observations included in the analysis. I will examine whether missing and suppressed QWI values are concentrated in certain industry types or states, and whether excluding smaller state-industry groups changes the results. Because AI exposure is measured at the industry level and observations from the same industry may be related across states and quarters, I will use industry-level clustered standard errors. As a sensitivity check, I will compare industry-clustered standard errors with alternative clustering specifications where the number of clusters permits reliable inference [10]. I will report coefficient estimates, confidence intervals, and statistical significance, while also considering the size and practical meaning of the estimated relationships.

An important limitation is that employment above the predicted pre-COVID trend is not a direct measure of overhiring or of the effect of the pandemic. The measure may reflect forecasting error, nonlinear pre-pandemic growth, changes in industry demand, regression to the mean, or continued COVID-related adjustment. COVID may also have affected industries through channels not captured by employment levels, including changes in demand, remote work, and labor supply. Including the employment-above-trend measure in the regression will therefore not fully separate AI-related differences from post-pandemic adjustment or other economic changes. My conclusions will focus on whether AI exposure remains associated with changes in early-career hiring after accounting for this measurable difference in earlier employment growth.

My primary analysis will use the fixed-effects panel regression described above, with the event-study specification serving as a secondary analysis of the timing of the estimated relationships. Alternatively, I will calculate the change in each state-industry group's average hiring rate between 2022 Q1-Q3 and 2023 Q1-Q4. I will then use a simpler regression to examine whether that change is associated with AI exposure and employment above the pre-COVID trend. This approach will require fewer modeling choices and may be useful if the quarterly panel is too incomplete for an event study. However, it will provide less information about the timing of hiring changes and whether differences were already developing before 2023. If four-digit industry data are too sparse, I will also consider using broader industry categories.

Reproducibility

I will conduct the data preparation and statistical analysis in Python [12]. I plan to use pandas to clean and combine datasets [13], statsmodels for regression, trend estimation, and model diagnostics [14], and PyFixest for fixed-effects estimation [15]. I will organize the project in a public GitHub repository containing the analysis scripts, a README explaining the data sources and steps needed to reproduce the analysis, and the final figures and tables. I will record the software and package versions used and keep the original data separate from cleaned datasets. I will share source links and download instructions for any datasets that cannot be redistributed.

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